

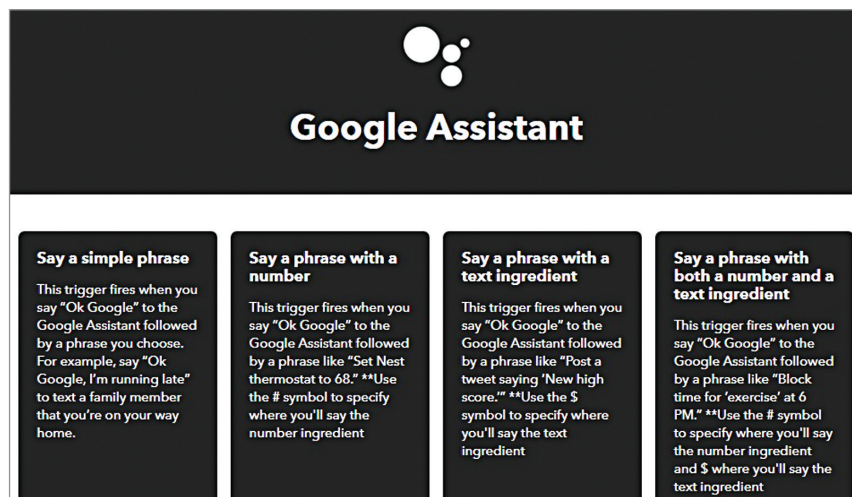


Fig. 7: Google Assistant on IFTTT platform

need to connect it to your Google account.

11. For relay control, you need to enter the following URL along with IP address of Blynk app, Auth Token, and GPIO pin in the text box: <https://188.166.206.43/your auth token/update/D2>

(D2 is the GPIO2 of ESP32). Here you need to paste in your Auth Token in the URL.

12. Next, select the Method field as 'PUT', Content type as 'Application/JSON' and Body type as ['1'] for turning on the light.

Similarly, you need to follow the same steps for turning on fan and TV.

13. For servo motor control, use the following URL:

<https://188.166.206.43/your auth token/update/V2>

(V2 is the virtual pin D23 of ESP32 for servo motor control defined in the source code and for position control ['1']). Here, V2 corresponds to 90-degree position (vertical) and V1 corresponds to 0-degree position (flat).

14. Next, select the Method field as 'PUT', Content type as 'Application/JSON' and Body type as ['1']

15. Now click on 'Create Action' and then on 'Finish.'

16. Next, create another applet in the same way to turn off the light/fan/TV. That is, you need to fill the same

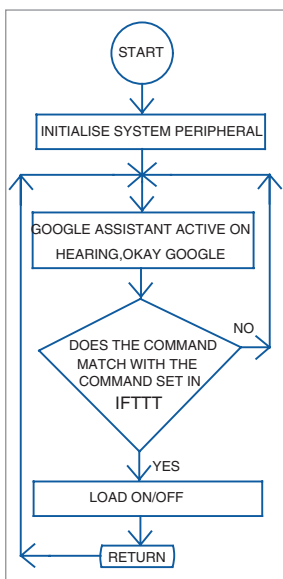


Fig. 8: System flowchart

data as above but with slight changes in the Body type. For turning off the device, the Body type should be ['0']

In the servomotor, the URL should be <https://188.166.206.43/your auth token/update/V1>

17. Hit 'Finish' to complete the software settings.

Next, activate Google Assistant app from your mobile phone, so that it is running in the background. Now, you can control the appliances either from your Blynk app with your finger or by just giving voice command to switch on/off the devices.

For servo motor, you can use voice command such as "bed up" for the servo motor to turn 90 degrees to lift the bed's backrest. When you say, "bed down," servo motor will turn to 0-degree position to bring the bed back to normal position.

Flowchart of the project is shown in Fig. 8 and the source code (*bedrid-den.ino*) is listed below:

```

Source code
#define BLYNK_PRINT Serial
#include <WiFi.h>
#include <WiFiClient.h>
#include <BlynkSimpleEsp32.h>
#include <Servo.h>

Servo servo;

// You should get Auth Token in the Blynk

```

PARTS LIST

Semiconductors:

- Board1 - ESP32 development board
- T1-T3 - BC547 NPN transistor
- D1-D3 - 1N4007 rectifier diode
- LED1-LED3 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1-R6 - 1-kilo-ohm resistor

Miscellaneous:

- RL1-RL3 - 5V single-changeover relay
- Servo motor - 3-pin micro servo motor
- CON1 - 2-pin terminal connector for 5V DC
- CON2 - 6-pin terminal connector from appliances
- CON3 - 3-pin connector for servo motor

```

App.
// Go to the Project Settings (nut icon).
char auth[] = "Your auth token";
// Your WiFi credentials.
// Set password to "" for open networks.
char ssid[] = "Your SSID";
char pass[] = "Your password";

void setup()
{
  Serial.begin(9600);
  Blynk.begin(auth, ssid, pass);
  servo.attach(23);
}

void loop()
{
  Blynk.run();
}

BLYNK_WRITE(V1)
{
  servo.write(0);
}

BLYNK_WRITE(V2)
{
  servo.write(90);
}

```

EFY



S. Maheshwaran is an electronics hobbyist. His areas of interest include automation, embedded systems, and robotics



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